NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics
Department of Computer Science

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COURSE OUTLINE: DATA NETWORK (DTN611S)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect, academic honesty and integrity is very important to ensure that learning is valid, reliable and credible.

NUST, therefore, does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or
- unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

COURSE INFORMATION

Course Code and Title: Data Networks – DTN611S

Department: Computer Science

Faculty: Faculty of Computing and Informatics

Programme: 07BACS – Bachelor of Computer Science (Systems Administration, Communication Networks, Software Development)

Contact Hours: **Total Hours: 120 hours**
Contact: 45 hours
Self-Directed Study: 55 hours
Assessments: 20 hours

NQF Level and Credit: NQF Level 6
12 Credits

Course Description: This course aims to provide students with an understanding of the underlying principles as well as the design, implementation and operation of both fixed and wireless networks. The course will also expose students to practical network environments through the configuration and troubleshooting of network topologies using network simulation tools. This is an exam-based course, and students are expected to obtain a minimum of 50% to qualify for the exam.

Pre-Requisites: None

Course Equivalent: None

Course Delivery Methods: The course will be facilitated through face-to-face sessions:

- Class lectures
- Class tutorials
- Network tools and simulators
- Self-Study
- Students' group discussions and presentations, projects, practical labs

Effective Date: 9 February 2026

Lecturer's Information:

Ms. Helena Hainana
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061-207 2099
IT House

Ms Jovita Matesu
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Mr. Joseph Rufuma
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Ms. Lineekela Shishiveni
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Ms. Taleni Andjamba
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Library – Lower Campus

Student Readiness

Technology & Equipment Readiness:

- This course will make use of existing technology within NUST. However, having your laptop will be helpful. Wireshark and Cisco Packet Tracer are extensive tools for this course and having access to them on your personal equipment will be beneficial.
- Students are encouraged to utilize network connections on and off-campus to access the course on our e-learning.

Student Commitments and Contact Times:

All students are expected to:

- Attend all theory and practical classes as scheduled on their timetables unless otherwise stated by the course coordinator. In addition, students are encouraged to take appropriate notes.
- Any changes to the time slots will be communicated to students by the course coordinator. To obtain and read through the lecture notes and other study materials before the commencement of the class.
- Avoid any disruptions (such as mobile phones, browsing the net, chatting to each other) during the class unless requested to do so by the lecturer.

- Lecturers are always ready to assist if you fail to understand the concepts. Therefore, make good use of the appointment slots outlined by each lecturer.

Course Resources:

- We will use computers in the IT labs.

Prescribed Textbooks:

- Kurose, J. & Ross, K. (2017). Computer networking: A top-down approach (7th ed.). Boston: Pearson.
- Peterson, L. & Davie, B. (2012). Computer networks: A systems approach. Amsterdam Boston: Morgan Kaufmann.

Recommended Learning Resources:

- Tanenbaum, A. (2010). Computer networks (5th ed.). Prentice Hall

Student Learning Outcomes

Upon completion of the course students will, through assessment activities, show evidence of their ability to:

- Examine how Internet is constructed and how it evolved, Identify and troubleshoot common network problems.
- Explain the various networking devices and their functions in enabling data communications.
- Describe the layered structure of a typical networked architecture (OSI and TCP/IP), and the common protocols, and the functions of each layer.
- Identify and classify different types of networks based on their characteristics and purposes.
- Demonstrate the ability to configure a basic network topology for a given scenario.
- Understand and demonstrate network addressing and subnet ting concept, including contrasting between IPv4 and IPv6.
- Illustrate how the transport layer supports end- to-end connectivity
- Illustrate how the application layer supports client-server communication.
- Explain how the data link layer ensures reliable communication, and the common methods used to address the multiple access problem.
- Differentiate between routing and forwarding.
- Explain the advantages, disadvantages and differences between wired and wireless networks.
- Understand the fundamentals of common network security threats and solutions.
- Demonstrate a deepened knowledge and understanding of the organisation of a wireless network and other emerging technologies.

Course Content

Week	Week Duration	Topic	Practical	Assessment
1	09 Feb – 13 Feb	Course Outline/ Course Overview		
2	16 Feb – 20 Feb	• Introduction to Computer Networks **LESSON1**		
3	23 Feb – 27 Feb	• Network Models and Standards **LESSON 2**		
4	02 Mar – 06 Mar	• Application Layer **LESSON 3**		
5	09 Mar – 13 Mar	• Transport Layer **LESSON 4**	Lab1	
6	16 Mar – 20 Mar	• Network Layer **LESSON 5**	Lab 2	
7	23 Mar – 27 Mar	• Data Link Layer **LESSON 6**	Conclude Lab 1 and Lab 2	**Test 1** **Date:** Saturday 28th March **Time:** 09h30 AM -11h30 AM **Venues:** 1.Engineering Basement SECTION A&B 2. Auditorium North & South **Scope:** Lesson 1 – Lesson 5
8	30 Mar – 03 Apr	**Mid-Semester Break**		
9	06 Apr – 10 Apr	• Physical Layer **LESSON 7**	Lab 3	
10	13 Apr – 17 Apr	• Lab Reviews	Lab 4	
11	20 Apr – 24 Apr	• Lab Reviews	Conclude Lab 3 and Lab 4	**Test 2** **Date:** Saturday 25th April

				Time: 08h00 AM – 10h00 AM **Venues:** 1.Engineering Basement SECTION A&B 2. Auditorium North & South **Scope:** Lab 1 – Lab 4
12	27 Apr – 01 May	• Lab Reviews	Lab Reviews	**Quiz** **Date:** Saturday 2nd May **Time:** To be confirmed **Scope:** Lesson 1 – Lesson 7 **E-Learning**
13	04 May – 08 May	• Lab Reviews	Lab Reviews	**Supp Test** **Date:** Saturday 9th May **Time:** 09:30 AM– 11:30 AM **Venue:** 1.Engineering Basement SECTION A&B **Scope:** Lesson 1 – Lesson 7 & Lab 1 – Lab 4
14	11 May – 12 May	• Lab/Mark Reviews	Lab/Marks Reviews	
15	13 May	• Lab reviews • **Lectures End for Exam Based**		
16	19 May	• **Last day to submit semester marks**		

Important Dates:

NOTE: The assessment dates are subject to change based on venue availability and other unforeseen circumstances at the lecturer's prerogative. Students will be notified during class, or on online platforms ahead of time of any changes.

Date	Information	Venue and Time
Saturday 28th March 2026	Test 1 Scope: Lesson 1 – Lesson 5	1.Engineering Basement SECTION A&B 2. Auditorium North & South 09h30 AM -11h30 AM
Saturday 25th April 2026	Test 2 Scope: Lab 1 – Lab 4	1.Engineering Basement SECTION A&B 2. Auditorium North & South 08h00 AM – 10h00 AM
Saturday 2nd May 2026	Quiz Scope: Lesson 1 – Lesson 7	Online: E-Learning Time to be Confirmed
Saturday 9th May 2026	Supplementary test Scope: Lesson 1 – Lesson 7 & Lab 1 – Lab 4	Engineering Basement SECTION A&B 09:30 AM– 11:30 AM

Note: The supplementary test will be written by students who missed one of the tests or those who did not qualify for the exams and would like to improve their semester marks to qualify for the exams. The supplementary will only replace either one of the test marks and not the lab assessments. If a student misses or obtains an unsatisfactory mark in any of the lab assessments, there will be no supplementary lab assessment to replace or improve their lab assessment semester marks.

Assessment and Evaluation

This is an exam-based course. Contribution to the final mark will be 40% Coursework and 60% written Final Examination. A minimum of 40% is required in the coursework mark for exam admission, and to pass the course a combined average of not less than 50% is required.

Assessment	Weight
Test 1	35%
Test 2	40%
Quiz	25%
Total Semester Mark	**100%**

Course Policies

General Academic Policies:
It is the student's responsibility to be familiar with and adhere to the NUST's Policies. These policies can be found in the NUST Prospectus or online at www.nust.na/prospectus.

Tests:

- The tests will be written as scheduled. The venue and time of the tests will be announced in advance.
- The tests only cover the material covered during the theory and practical classes.

Class attendance:

- Students are required to attend all classes. Test admission may be withheld if a student does not attend at least 80% of classes.
- Students need to participate in all assessments and tests.

Date Revised:

Failure to Pay Fees:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full, subjected to the Policy of the University.

Important Student Services at NUST

There are a variety of services that you can use at the NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development - Dean of Students Office.
- Writing centre and student academic problems – Centre for Teaching & Learning (CTL)
- Campus Health and Wellness Centre (CHWC) - Dean of Student's office/ NUSTClinic
- University Library

Authorization:

This course is authorized for use by:

_____ _____

Head of Department Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I,…….. (insert name), (Student number)……, hereby acknowledge that I have received this course outline for ……..(insert course title and code), and that I have familiarized myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of Student _____